Prayas JEE AIR (2025)

WEEKLY TEST 24

DURATION : 180 Minutes | **DATE : 26/12/2024** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ Invigilator's Signature _____

SECTION-I (PHYSICS)

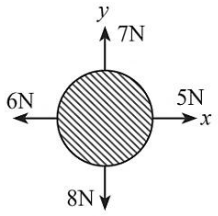
Single Correct Type Questions

1. A ball is thrown up vertically with a certain velocity at $t = 0$ so that, it reaches a maximum height h. Find the ratio of the time instants in which it is at height $\frac{h}{3}$ while going up and coming down respectively.

(1) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (2) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(3) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (4) $\frac{1}{3}$

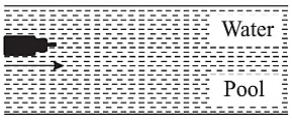
2. For a free body diagram shown in the figure, the four forces are applied in the 'x' and 'y' directions. What additional force must be applied and at what angle with positive x-axis so that the net acceleration of body is zero?



(1) $\sqrt{2}\,\text{N}, 45^\circ$ (2) $\sqrt{2}\,\text{N}, 135^\circ$

(3) $\frac{2}{\sqrt{3}}\,\text{N}, 30^\circ$ (4) $2\,\text{N}, 45^\circ$

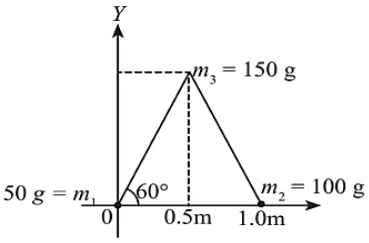
3. A bullet of mass 200g having initial kinetic energy 90 J is shot inside a long swimming pool as shown in the figure. If it's kinetic energy reduces to 40 J within 1s, the minimum length of the pool, the bullet has to travel so that it completely comes to rest is



(1) 45 m (2) 90 m

(3) 125 m (4) 25 m

4. Three particles of masses 50 g, 100 g and 150 g are placed at the vertices of an equilateral triangle of side 1 m (as shown in the figure). The (x, y) coordinates of the centre of mass will be :



(1) $\left(\frac{\sqrt{3}}{7}\text{m}, \frac{7}{12}\text{m}\right)$ (2) $\left(\frac{7}{12}\text{m}, \frac{\sqrt{3}}{8}\text{m}\right)$

(3) $\left(\frac{\sqrt{3}}{4}\text{m}, \frac{5}{12}\text{m}\right)$ (4) $\left(\frac{7}{12}\text{m}, \frac{\sqrt{3}}{4}\text{m}\right)$

5. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): Moment of inertia of a circular disc 'M' and radius 'R' about X, Y axes (passing through its plane) and Z-axis which is perpendicular to its plane were found to be I_x, I_y and I_z respectively. The respective radii of gyration about all the three axes will be the same.

Reason (R): A rigid body making rotational motion has fixed mass and shape. In the light of the above statements, choose the most appropriate answer from the options given below:

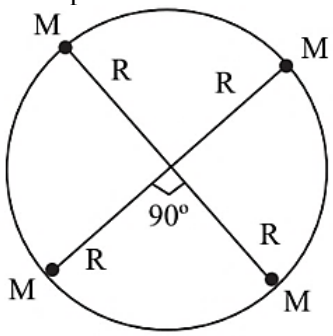
(1) A is correct but R is not correct.

(2) Both A and R are correct and R is the correct explanation of A.

(3) Both A and R are correct but R is NOT the correct explanation of A.

(4) A is not correct but R is correct.

6. Four particles each of mass M, move along a circle of radius R under the action of their mutual gravitational attraction as shown in figure. The speed of each particle is:



(1) $\frac{1}{2}\sqrt{\frac{GM}{R(2\sqrt{2}+1)}}$

(2) $\frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}-1)}$

(3) $\frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}+1)}$

(4) $\sqrt{\frac{GM}{R}}$

7. A force is applied to a steel wire 'A', rigidly clamped at one end. As a result elongation in the wire is 0.2 mm. If same force is applied to another steel wire 'B' of double the length and a diameter 2.4 times that of the wire 'A', the elongation in the wire 'B' will be (wires having uniform circular cross sections)

(1) 6.06×10^{-2} mm (2) 2.77×10^{-2} mm

(3) 3.0×10^{-2} mm (4) 6.9×10^{-2} mm

8. Match Column-I with Column -II

Column-I		Column-II	
A	Torque	P	$ML^{-2}T^{-2}$
B	Stress	Q	$ML^{2}T^{-2}$
C	Pressure gradient	R	$ML^{-1}T^{-1}$
D	Coefficient of viscosity	S	$ML^{-1}T^{-2}$

Choose the correct answer from the options given below:

(1) A→R, B→S, C→P, D→Q

(2) A→S, B→Q, C→R, D→P

(3) A→Q, B→S, C→P, D→R

(4) A→Q, B→P, C→S, D→R

9. A hypothetical gas expands adiabatically such that its volume changes from 8 litres to 27 litres. If the ratio of final pressure of the gas to initial pressure of the gas is $\frac{16}{81}$. Then the ratio of $\frac{C_P}{C_V}$ will be.

(1) $\frac{4}{3}$ (2) $\frac{3}{1}$

(3) $\frac{1}{2}$ (4) $\frac{3}{2}$

10. Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance (R/2) from the earth's centre, where 'R' is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period :

(1) $2\pi\sqrt{\frac{R}{g}}$ (2) $2\pi\sqrt{\frac{2R}{g}}$

(3) $\frac{\pi}{2}\sqrt{\frac{g}{R}}$ (4) $2\pi\sqrt{\frac{R}{2g}}$

11. Two identical tennis balls each having mass 'm' and charge 'q' are suspended from a fixed point by threads of length 'l'. What is the equilibrium separation when each thread makes a small angle 'θ' with the vertical?

(1) $x=\left(\frac{q^2\ell}{2\pi\varepsilon_0 mg}\right)^{1/2}$

(2) $x=\left(\frac{q^2\ell}{2\pi\varepsilon_0 mg}\right)^{1/3}$

(3) $x=\left(\frac{q^2\ell^2}{2\pi\varepsilon_0 m^2 g^2}\right)^{1/3}$

(4) $x=\left(\frac{q^2\ell^2}{2\pi\varepsilon_0 m^2 g}\right)^{1/3}$

12. Ten charges are placed on the circumference of a circle of radius R with constant angular separation between successive charges. Alternate charges 1, 3, 5, 7, 9 have charge ($+q$) each, while 2, 4, 6, 8, 10 have charge ($-q$) each. The potential V and the electric field E at the center of the circle are respectively. (Take $V = 0$ at infinity)

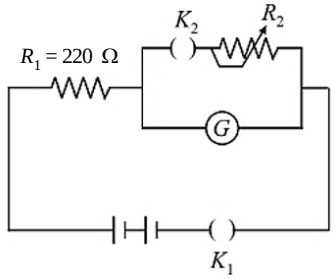
(1) $V=0; E=0$

(2) $V=\frac{10q}{4\pi\varepsilon_0 R}; E=\frac{10q}{4\pi\varepsilon_0 R^2}$

(3) $V=\frac{10q}{4\pi\varepsilon_0 R}=E=0$

(4) $V=0; E=\frac{10q}{4\pi\varepsilon_0 R^2}$

13. The galvanometer deflection, when key K_1 is closed but K_2 is open, equals θ_0 (see figure). On closing K_2 also and adjusting R_2 to 5 Ω, the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by [Neglect the internal resistance of battery]:



(1) 5 Ω (2) 22 Ω

(3) 25 Ω (4) 12 Ω

14. The magnetic field at the centre of a circular coil of radius r, due to current I flowing through it, is B. The magnetic field at a point along the axis at a distance r/2 from the centre is:

(1) $\frac{B}{2}$ (2) $2B$

(3) $\left(\frac{2}{\sqrt{5}}\right)^3 B$ (4) $\left(\frac{1}{\sqrt{3}}\right)^3 B$

15. A long solenoid of radius R carries a time (t) dependent current $I(t) = I_0 t(1-t)$. A ring of radius $2R$ is placed cordially near its middle. During the time interval $0 \le t \le 1$, the induced current (I_R) and the induced emf (V_R) in the ring change as:

(1) Direction of I_R remains unchanged and V_R is zero at $t = 0.25$

(2) At $t = 0.5$ direction of I_R reverses and V_R is zero

(3) At $t = 0.25$ direction of I_R reverses and V_R is maximum

(4) Direction of I_R remains unchanged and V_R is maximum at $t = 0.5$

16. A convex lens (of focal length 20 cm) and a concave mirror, having their principal axes along the same lines, are kept 80 cm apart from each other. The concave mirror is to the right of the convex lens. When an object is kept at a distance of 30 cm to the left of the convex lens, its image (formed by the concave mirror after reflection only) will form at the same position even if the concave mirror is removed. The maximum distance of the object (from mirror) for which this concave mirror, by itself would produce a virtual image would be:

(1) 20 cm (2) 10 cm
(3) 25 cm (4) 30 cm

17. In a double slit experiment, when a thin film of thickness t having refractive index μ is introduced in front of one of the slits. The central maximum of the fringe pattern shifts by one fringe width. The value of t is (λ is the wavelength of the light used)

(1) $\frac{\lambda}{2(\mu-1)}$

(2) $\frac{\lambda}{(2\mu-1)}$

(3) $\frac{2\lambda}{(\mu-1)}$

(4) $\frac{\lambda}{(\mu-1)}$

18. An electromagnetic wave is represented by the electric field $\vec{E}=E_0\hat{n}\sin[\omega t+(6y-8z)]$. Taking unit vectors in x, y and z directions to be $\hat{i}$, $\hat{j}$, $\hat{k}$, the direction of propagation $\hat{s}$ is

(1) $\hat{s}=\left(\frac{4\hat{j}-3\hat{k}}{5}\right)$

(2) $\hat{s}=\left(\frac{-3\hat{i}-4\hat{j}}{5}\right)$

(3) $\hat{s}=\left(\frac{-3\hat{j}+4\hat{k}}{5}\right)$

(4) $\hat{s}=\left(\frac{-4\hat{k}+3\hat{j}}{5}\right)$

19. A particle of mass 200 MeV/c^2 collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is $\frac{N}{4}$. The value of N is: (Given the mass of the hydrogen atom to be 1 GeV/c^2)

(1) 25 (2) 102
(3) 51 (4) 50

20. A free neutron decays into a proton but a free proton does not decay into neutron. This is because

(1) neutron is an uncharged particle
(2) proton is a charged particle
(3) neutron is a composite particle made of a proton and an electron
(4) neutron has larger rest mass than proton

Integer Type Questions

21. A string of length 0.4 m and mass 10^{-2} kg is clamped at its ends. The tension in the string is 1.6 N. Identical wave pulses are generated at one end at regular intervals of time, Δt. The minimum value of Δt, so that a constructive interference takes place between successive pulses is_____(in ms).

22. The electric field in a region is given by $\vec{E}=(Ax+B)\hat{i}$ where E is in NC^{-1} and x is in metres. The values of constants are A = 20 *SI* unit and B = 10 *SI* unit. If the potential at x = 1 is V_1 and that at x = –5 is V_2, then $V_1 - V_2$ (in volt) is ____.

23. An inductor of inductance 100 mH is connected in series with a resistance, a variable capacitance and an *AC* source of frequency $\frac{2}{\pi}$KHz. What should be the value of the capacitance (in nF) so that maximum current may be drawn into the circuit ?

24. If the de-Broglie wavelength of an electron is equal to 663 times the wavelength of a photon of frequency 27.3×10^{14} Hz, then the speed (in m/s) of electron is equal to : (Speed of light = 3×10^8 m/s)
Planck's constant = 6.63×10^{-34}J.s
Mass of electron = 9.1×10^{-31} kg)

25. M grams of steam at 100°*C* is mixed with 200 *g* of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°*C* [heat of vaporization of water is 540 cal/*g* and heat of fusion of ice is 80 cal/*g*], the value of *M* is .

SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. Consider the following pairs of electrons

(A) (a) $n=3, l=1, m_\ell=1, m_s=+\frac{1}{2}$

(b) $n=3, l=2, m_\ell=1, m_s=+\frac{1}{2}$

(B) (a) $n=3, l=2, m_\ell=-2, m_s=-\frac{1}{2}$

(b) $n=3, l=2, m_\ell=-1, m_s=-\frac{1}{2}$

(C) (a) $n=4, l=2, m_\ell=2, m_s=+\frac{1}{2}$

(b) $n=3, l=2, m\ell=-2, m_s=+\frac{1}{2}$

The pairs of electrons present in degenerate orbitals is/are:

(1) Only (A) (2) Only (B)
(3) Only (C) (4) (B) and (C)

27. Trichloroacetaldehyde was subjected to Cannizzaro's reaction by using NaOH. The mixture of the products contains sodium trichloroacetate and another compound. The other compound is:

(1) trichloromethanol
(2) 2, 2, 2-trichloropropanol
(3) chloroform
(4) 2, 2, 2-trichloroethanol

28. Given below are two statements :

Statement-I: SO_2 and H_2O both possess V-shaped structure.

Statement-II: The bond angle of SO_2 is less than that of H_2O

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct
(2) Statement-I is correct but Statement-II is incorrect
(3) Both Statement-I and Statement-II are incorrect
(4) Statement-I is incorrect but Statement-II is correct

29. Fructose is an example of:

(1) Pyranose (2) Ketohexose
(3) Aldohexose (4) Heptose

30. Consider the reaction:

$Cl_2(aq) + H_2S(aq) \rightarrow S(s) + 2H^+(aq) + 2Cl^-(aq)$

The rate equation for this reaction is rate =k[Cl_2] [H_2S]. Which of these mechanisms is/are consistent with this rate equation?

(i) $Cl_2 + H_2S \rightarrow H^+ + Cl^- + Cl^+ + HS^-$ (slow)
$Cl^+ + HS^- \rightarrow H^+ + Cl^+ + S$ (fast)

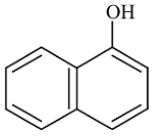
(ii) $H_2S \rightleftharpoons H^+ + HS^-$ (faste quilibrium)
$Cl_2 + HS^- \rightarrow 2Cl^- + H^+ + S$ (slow)

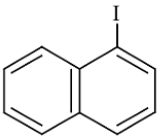
(1) (ii) only
(2) Both (i) and (ii)
(3) Neither (i) nor (ii)
(4) (i) only

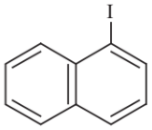
31. If K_{sp} of Ag_2CO_3 is 8×10^{-12}, the molar solubility of Ag_2CO_3 in 0.1M $AgNO_3$ is:

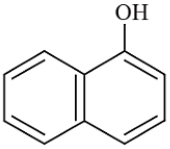
(1) 8×10^{-12} M (2) 8×10^{-11} M
(3) 8×10^{-10} M (4) 8×10^{-13} M

32. Main products formed during a reaction of 1-methoxy naphthalene with hydroiodic acid are:

(1)  and CH_3OH

(2)  and CH_3I

(3)  and CH_3OH

(4)  and CH_3I

33. If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water (ΔT_f), when 0.01 mol of sodium sulphate is dissolved in 1 kg of water, is
(K_f =1.86 K kg mol^{-1})

(1) 0.0372 K (2) 0.0558 K
(3) 0.0744 K (4) 0.0186 K

34. The spin only magnetic moments (in BM) for free Ti^{3+}, V^{2+} and Sc^{3+} ions respectively are (At. No. Sc: 21; Ti: 22; V: 23)

(1) 1.73, 0, 3.87 (2) 0, 3.87, 1.73
(3) 3.87, 1.73, 0 (4) 1.73, 3.87, 0

35. Match Column-I with Column-II.

	Column-I (Parameter)		**Column -II (Unit)**
A	Cell constant	P	Scm^2mol^{-1}
B	Molar conductivity	Q	Dimensionless
C	Conductivity	R	m^{-1}
D	Degree of dissociation of electrolyte	S	$\Omega^{-1} m^{-1}$

Choose the most appropriate answer from the options given below:

(1) A→R; B→ P; C→Q; D→S
(2) A→Q; B→ P; C→R; D→S
(3) A→R; B→P; C→S; D→Q
(4) A→P; B→S; C→R; D→Q

36. $C(s) + O_2(g) \rightarrow CO_2(g) + 400\,kJ$

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g) + 100\,kJ$

When coal of purity 60% is allowed to burn in presence of insufficient oxygen, 60% of carbon is converted into CO and the remaining is converted into CO_2. The heat generated when 0.6 kg of coal is burnt is_____.

(1) 1600 kJ (2) 3200 kJ
(3) 4400 kJ (4) 6600 kJ

37. For the reaction:

$$RCH_2Br + I^- \xrightarrow[\text{majjor}]{\text{Acetone}} RCH_2I + Br^-$$

The correct statement is:

(1) The transition state formed in the above reaction is less polar than the localised anion.
(2) The reaction can occur in acetic acid also.
(3) The solvent used in the reaction solvates the ions formed in rate determining step.
(4) Br^- can act as competing nucleophile.

38. The exact volumes of 1 M NaOH solution required to neutralize 50 mL. of $1MH_3PO_3$ solution and 100 mL of $2MH_3PO_2$ solution respectively, are:

(1) 100 mL and 50 mL
(2) 100 mL and 200 mL
(3) 50 mL and 50 mL
(4) 100 mL and 100 mL

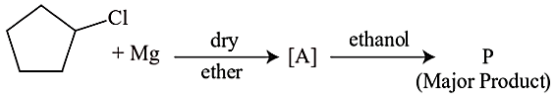
39. Three complexes, $[CoCl(NH_3)_5]^{2+}$ (I), $[Co(NH_3)_5 H_2O]^{3+}$ (II) and $[Co(NH_3)_6]^{3+}$ (III) absorb light in the visible region. The correct order of the wavelength of light absorbed by them is:

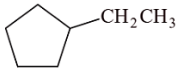
(1) (III) > (I) > (II) (2) (I) > (II) > (III)
(3) (II) > (I) > (III) (4) (III) > (II) > (I)

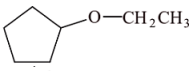
40. Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is:

(1) X < Z < Y (2) Z < Y < X
(3) X < Y < Z (4) Y < X < Z

41. In the following sequence of reactions, the P is:



(1) 

(2) 

(3) 
(4) 

42. Given below are two statements:

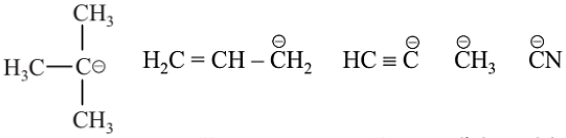
Statement-I: Chlorine can easily combine with oxygen to from oxides, and the product has a tendency to explode.

Statement-II: Chemical reactivity of an element can be determined by its reaction with oxygen and halogens.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both the statements-I and Statement-II are true
(2) Statement-I is true but Statement-II is false
(3) Statement-I is false but Statement-II is true
(4) Both the Statements-I and II are false

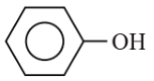
43. The increasing order of basicity for the following intermediates is (from weak to strong)

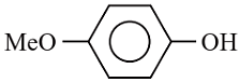


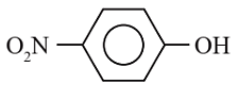
(i) (ii) (iii) (iv) (v)

(1) (v) < (iii) < (ii) < (iv) < (i)
(2) (iii) < (i) < (ii) < (iv) < (v)
(3) (v) < (i) < (iv) < (ii) < (iii)
(4) (iii) < (iv) < (ii) < (i) < (v)

44. The correct order for acidity of the following hydroxyl compound is:

(A) CH_3OH (B) $(CH_3)_3COH$
(C) 

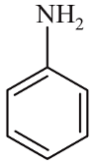
(D) 

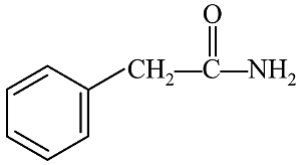
(E) 

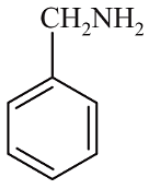
Choose the correct answer from the options given below:

(1) E > C > D > A > B
(2) E > D > C > A > B
(3) C > E > D > A > B
(4) E > D > C > B > A

45. Which of the following compounds can be prepared in good yield by Gabriel phthalimide synthesis?

(1) 

(2) 

(3) 

(4) $CH_3– CH_2– NHCH_3$

Integer Type Questions

46. $2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(s)$

This reaction was studied at -10°C and the following data was obtained.

run	$[NO]_0$	$[Cl_2]_0$	r_0
1.	0.10	0.10	0.18
2.	0.10	0.20	0.35
3.	0.20	0.20	1.40

$[NO]_0$ and $[Cl_2]_0$ are the initial concentrations and r_0 is the initial reaction rate. The overall order of the reaction is_____.
(Round off to the nearest integer).

47. 1 L aqueous solution of H_2SO_4 contains 0.02 m mol H_2SO_4. 50% of this solution is diluted with deionized water to give 1 L solution (A). In solution (A), 0.01 m mol of H_2SO_4 are added. Total m mols of H_2SO_4 in the final solution is_____ $\times 10^3$ m moles. (Nearest Integer)

48. The magnetic moment (spin only) of $[NiCl_4]^{2-}$ is___ BM (Nearest Integer)

49. In Tollen's test for aldehyde, the overall number of electron(s) transferred to the Tollen's reagent formula $[Ag(NH_3)_2]^+$) per aldehyde group to form silver mirror is _____.
(Round off to the nearest integer).

50. Mass of Urea (NH_2CONH_2) required to be dissolved in 1000 g of water to reduce the vapour pressure of water by 25% is _____ Kg. (Nearest Integer)
Given: Molar mass of N, C, O and H are 14, 12, 16 and 1g mol^{-1} respectively.

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ If $a = (1+\alpha)\sum_{k=0}^{100}\alpha^{2k}$ and $b = \sum_{k=0}^{100}\alpha^{3k}$,

The equation whose roots are *a* and *b* is

(1) $x^2 + 101x + 100 = 0$
(2) $x^2 – 102x + 101 = 0$
(3) $x^2 – 101x + 100 = 0$
(4) $x^2 + 102x + 101 = 0$

52. Let $A = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$. If $B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} A \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50} B^n$ is equal to

(1) 100 (2) 50
(3) 75 (4) 125

53. If $C_r = {}^{25}C_r$ and $C_0 + 5\cdot C_1 + 9\cdot C_2 + \ldots + (101)\cdot C_{25} = 2^{25}\cdot k$, then *k* is equal to ____

(1) 49 (2) 50
(3) 51 (4) 52

54. Let z_1 and z_2 be any two non-zero complex such that $3|z_1| = 4|z_2|$. If $z = \frac{3z_1}{2z_2} + \frac{8z_2}{3z_1}$ then;

(1) Re (*z*) = 0 (2) $z_1 = z_2$
(3) $|z_1| = z_2$ (4) Im (*z*) = 0

55. Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2 + 11a + 3$ $\left(m_1^2 + m_2^2\right) = 220$. If one vertex of the square is (10(cos α – sin α), 10 (sin α + cos α)), where $\alpha \in \left(0, \frac{\pi}{2}\right)$ and the equation of one diagonal is (cos α – sin α)*x* + (sin α + cos α) *y* = 10, then $m_1^2 + m_2^2$ equals

(1) 3 (2) 10/3
(3) 10 (4) 3/10

56. The distance of the point *P* (4, 6, – 2) from the line passing through the point (–3, 2, 3) and parallel to *a* line with direction ratios <3, 3, –1> is equal to:

(1) 3 (2) $\sqrt{6}$
(3) $2\sqrt{3}$ (4) $\sqrt{14}$

57. Consider the data on x taking the values 0, 2, 4. 8, …2^n with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, \ldots, {}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$ then n is equal to _____.

(1) 3 (2) 4
(3) 5 (4) 6

58. Let A_1, A_2, A_3 be the three A.P. with the same common difference d and having their first terms as $A, A+1, A+2$, respectively. Let a, b, c be the 7th, 9th, 17th terms of A_1, A_2, A_3 respectively such that $\begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0$ If a = 29, then the sum of first 20 term of AP whose first term is $c - a - b$ and common difference is $\frac{d}{12}$, is equal to

(1) 495 (2) 555
(3) 725 (4) 625

59. If the equation of the parabola, whose vertex is at (5,4) and the directrix is $3x + y - 29 = 0$, is $x^2 + ay^2 + bxy + ex + dy + k = 0$, then a is equal to

(1) 0 (2) 3
(3) 6 (4) 9

60. The number of point (s) where the function $f : R \to R, f(x) = |x-1|\cos|x-2|$ is non-differentiable, is

(1) 1 (2) 2
(3) 3 (4) 4

61. The total number of three-digit numbers, divisible by 3, which can be formed using the digits 1,3,5,8, if repetition of digits is allowed, is:

(1) 22 (2) 18
(3) 21 (4) 20

62. The value of $\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$ is

(1) $3\pi - 11$ (2) $3\pi + 1$
(3) $4\pi - 11$ (4) $4\pi - 9$

63. The complete set of values of x for which $f(x) = \ln(x^2 - 7x + 12)$ is always increasing is

(1) $\left(3, \frac{7}{2}\right) \cup (4, \infty)$ (2) R
(3) $\left(3, \frac{7}{2}\right)$ (4) $(4, \infty)$

64. Number of matrices having 9 elements can be formed using 0, 1, –1 as elements is

(1) 3^9 (2) 3^{10}
(3) 1 (4) None of these

65. An arc PQ of a circle subtends a right angle at its centre O. The mid-point of the arc PQ is R. If $\overrightarrow{OP} = \vec{u}, \overrightarrow{OR} = \vec{v}$ and $\overrightarrow{OQ} = \alpha\vec{u} + \beta\vec{v}$, then α, β^2 are the roots of the equation ($\beta > 0$)

(1) $x^2 - x - 2 = 0$ (2) $3x^2 + 2x - 1 = 0$
(3) $x^2 + x - 2 = 0$ (4) $3x^2 - 2x - 1 = 0$

66. Let $a > 0$, $b > 0$. Let e and ℓ respectively be the eccentricity and length of the latus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Let e' and ℓ' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2 = \frac{11}{14}\ell$ and $\left(e'\right)^2 = \frac{11}{8}\ell'$, then the value of $4b/a$ is equal to:

(1) 2 (2) 3
(3) 5 (4) 7

67. Let, $y = f(x)$ be the solution of the differential equation $y(x+1)dx - x^2dy = 0, y(1) = e.$
Then $\lim\limits_{x \to 0^+} f(x)$ is equal to

(1) 0 (2) $\frac{1}{e}$
(3) e^2 (4) $\frac{1}{e^2}$

68. The area of the region: $R = \{(x, y) : 5x^2 \le y \le 2x^2 + 9\}$ is:

(1) $9\sqrt{3}$ square units
(2) $12\sqrt{3}$ square units
(3) $6\sqrt{3}$ square units
(4) $11\sqrt{3}$ square units

69. A water tank has the shape of an inverted right circular cone, whose semi-vertical angle is $\tan^{-1}\left(\frac{1}{2}\right)$. Water is poured into it at a constant rate of 5 cubic meter per minute.
Then the rate (in m/min.), at which the level of water is rising at the instant when the height of water in the tank is 10 m; is:

(1) $2/\pi$ (2) $1/5\pi$
(3) $1/10\pi$ (4) $1/15\pi$

70. Let the sum of the coefficient of the first three terms in the expansion of $\left(x-\frac{3}{x^2}\right)^n, x \neq 0, n \in N$, be 376. Then the coefficient of x^4 is

(1) 495 (2) 555

(3) 405 (4) Zero

Integer Type Questions

71. If $\int_0^1 \left(x^{21}+x^{14}+x^7\right)\left(2x^{14}+3x^7+6\right)^{1/7} dx = \frac{1}{\ell}(11)^{m/n}$ where $\ell, m, n \in \mathbb{N}, m$ and n are coprime then $\ell + m + n$ is equal to

72. If one of the diameters of the circle $x^2+y^2-2\sqrt{2}x-6\sqrt{2}y+14=0$ is a and chord of circle. $(x-2\sqrt{2})^2+(y-2\sqrt{2})^2=r^2$, then the value of r^2 is equal to.

73. Let two non-empty sets A & B has m & n elements respectively, if number of subsets of A are 112 more than those of B then $4m + 3n$ is

74. Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the probability that a randomly chosen onto function g from S to S satisfies $g(3) = 2g(1)$ is $\frac{1}{k}$, then k is

75. The value of $\frac{3}{2}+\cos\left(\frac{2\pi}{7}\right)+\cos\left(\frac{4\pi}{7}\right)+\cos\left(\frac{6\pi}{7}\right)$ is equal to:



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if